

Consider the affine function

$$f : \mathbb{R}^n \rightarrow \mathbb{R} : x \mapsto c^T x + b = \sum_{i=1}^n c_i x_i + b,$$

where  $c \in \mathbb{R}^n$  and  $b \in \mathbb{R}$ . For what values of  $x$ ,  $b$  and  $c$  are the necessary optimality conditions verified?